

2. LIMITATIONS

2.1 SYSTEM INSTALLATION REQUIREMENTS

A magnetic compass deviation card calculated for emergency DC-bus operation shall be installed.

NOTE The compass deviation card for emergency DC-bus operation is not required when a slaved gyroscopic direction indicator with flux valve is installed.

3. EMERGENCY AND MALFUNCTION PROCEDURES

3.1 WARNING LIGHT INDICATIONS

WARNING LIGHT INDICATIONS

GEN 1

and

GEN 2

Conditions/Indications

Both generators have failed or are disconnected from the power distribution circuit.

Procedure

1. Each STARTER/GENERATOR switch – Off, then GEN ON in turn

If both **GEN** warning lights remain on:

2. EMERG DC-BUS pushbutton – Press; **EMERG DC-BUS** caution light legend comes on
3. Electrical consumption – Reduce
4. Voltmeter – Monitor
5. LAND AS SOON AS PRACTICABLE

NOTE Pressing the EMERG DC-BUS pushbutton causes both generators to go off-line; only equipment connected to the emergency bus is then supplied with battery power.

3.2 CAUTION LIGHT INDICATIONS

CAUTION LIGHT INDICATIONS

EMERG DC-BUS

Conditions/Indications

The EMERG DC-BUS pushbutton is pressed with the following results:

- a) only the emergency DC power distribution bus equipment is supplied power, and
 - b) power is supplied only from the onboard battery.
- **EMERG DC-BUS** caution light legend on.

Procedure

LAND AS SOON AS PRACTICABLE

Residual Battery Endurance					
Continuous load [A]	15	20	25	30	40
Time [min]	60	45	35	30	22
NOTE Calculations are based on an assumed minimum battery capacity of 15 Ah. Times include 10-min landing light operation and 10-min radio transmitting.					

NOTE • Emergency bus current is not indicated on the ammeter.

- Under normal conditions the battery capacity allows for approx 30-min operation. However, emergency battery-power operation may be prolonged by observing the following rules:
 - operate the forward main tank fuel pump only long enough to fill the supply tank,
 - dim the instrument lighting as much as conditions permit,
 - the landing light should only be used when absolutely necessary,
 - the stability augmentation system (SAS), if installed, should only be used when absolutely necessary.

3.3 SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

EFFECTIVITY 0007-0080

3.3.1 DC-bus Failure

Conditions/Indications

Short circuit of a primary bus possibly causing excessive current

- Voltmeter indication abnormal (below 27V or above 29V)
- Ammeter may indicate excessive current (pointer either full left or right)
- Failure of equipment powered by affected bus
- Several circuit breakers may trip
- Odor of burning electrical insulation

Procedure

1. EMERG DC-BUS pushbutton
 - Press; **EMERG DC-BUS** caution light legend on
2. Electrical consumption
 - Reduce
3. Voltmeter
 - Monitor
4. LAND AS SOON AS PRACTICABLE

CAUTION IF THE SHORT CIRCUIT INDICATIONS REMAIN AFTER PRESSING THE EMERG DC-BUS PUSHBUTTON, THE EMERGENCY BUS IS MOST LIKELY SHORT CIRCUITED AND THEREFORE ALL ELECTRICAL POWER MUST BE SHUT OFF.

If short circuit indications continue:

5. EMERG DC-BUS pushbutton
 - Release; **EMERG DC-BUS** caution light legend off
6. Battery switch
 - Off
7. LAND AS SOON AS POSSIBLE

NOTE Individual electrical circuits/equipment are protected against shorting by either a fuse or a circuit breaker. The above procedure deals with a major short circuit occurring within the electrical power distribution system.

EFFECTIVITY 0081-9999

3.3.2 Main Bus Failure

Conditions/Indications

Short circuit of a primary bus possibly causing excessive current

- Voltmeter indication abnormal (below 27V or above 29V)
- Ammeter may indicate excessive current (pointer either full left or right)
- Failure of equipment powered by affected bus
- Several circuit breakers may trip
- Odor of burning electrical insulation

Procedure

- | | |
|--------------------------------|--|
| 1. EMERG DC-BUS pushbutton | – Press; EMERG DC-BUS caution light legend on |
| 2. Electrical consumption | – Reduce |
| 3. Voltmeter | – Monitor |
| 4. LAND AS SOON AS PRACTICABLE | |

NOTE Individual electrical circuits/equipment are protected against shorting by either a fuse or a circuit breaker. The above procedure deals with a major short circuit occurring within the electrical power distribution system.

3.3.3 Emergency Bus Failure

Conditions/Indications

Short circuit of emergency bus possibly causing excessive current

- Voltmeter indication abnormal (below 27V or above 29V)
- Ammeter may indicate excessive current (pointer either full left or right)
- Failure of equipment powered by emergency bus (except those protected by RADIO NAV 1 circuit breaker)

Procedure

1. LAND AS SOON AS PRACTICABLE

NOTE The electrical system is protected by a fuse that will disconnect the emergency bus should a short circuit occur there.

EFFECTIVITY ALL